CariSurg MedTech Pathways Research Project

Project Title: Evaluating Machine Learning Models for Emergency Department Triage Using Physiological Vital Signs at Mercer General Hospital

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Statement of the Problem:

Across the literature, emergency departments face increasing pressure from overcrowding, rising patient volumes, and limited healthcare resources, making accurate and timely triage difficult. Traditional triage methods rely heavily on clinician judgment and standardized protocols, which may lead to inconsistencies in patient prioritization. Studies consistently show that artificial intelligence can improve triage accuracy, predict patient outcomes, and support healthcare decision-making (Tyler et al., 2024; Da'Costa et al., 2025). However, concerns regarding transparency, bias, reliability, clinician trust, and integration into existing workflows remain unresolved (Da'Costa et al., 2025; Araouchi & Adda, 2024). The existing literature focuses mainly on individual model performance, conceptual frameworks or reviews of AI-assisted triage systems. There remains limited comparative evaluation of baseline and ensemble machine learning models using physiological vital signs under controlled conditions and the robustness and reliability of these approaches have not been adequately investigated. Therefore there is a need for further research to compare machine learning models for triage prediction and provide evidence that may support further clinical implementation.

Previous Work:

Tyler et al. (2024) examines how artificial intelligence is used in emergency department triage to improve patient prioritization and resource allocation. The authors conducted a scoping review of 29 studies published between 2018 and 2023 focusing on AI and machine learning applications in triage. Findings showed that AI models often outperform traditional triage systems in predicting hospitalization, critical illness, and ICU admission. However, limitations include concerns about algorithm transparency, data quality, ethical issues, and the need for broader validation across different healthcare settings.

Da'Costa et al. (2025) explores the benefits, challenges, and future directions of AI-driven triage systems in emergency departments facing overcrowding and inconsistent patient prioritization. The authors synthesized existing literature on AI-assisted triage technologies and their impact on emergency care. Results suggest that AI can improve triage accuracy, support faster decision-making, and enhance resource management in high-demand environments. However, barriers such as data bias, system integration, explainability, regulatory compliance, and clinician trust continue to limit widespread adoption.

Araouchi and Adda (2024) focuses on overcrowding in emergency departments and the need for improved triage decision-making. The authors developed and evaluated several machine learning models, including Support Vector Machines, Random Forests, Gradient Boosting, XGBoost, and a stacking model, using the Korean Triage and Acuity Scale (KTAS). The stacking model achieved the best performance, with approximately 80% accuracy in predicting triage levels. A key limitation is that the model was tested on a specific dataset and triage framework, requiring further validation across different healthcare environments before general use.

Kim et al. (2023) examines the evolution of clinical decision support systems from traditional rule-based and machine learning approaches to explainable artificial intelligence (XAI) based systems on healthcare. It was highlighted that the knowledge-based CDSSs are very interpretable but limited by their high maintenance and data standardization costs. To address this, the paper proposes an XAI-based CDSS framework that integrates explainability techniques (e.g., SHAP, LIME, Grad-CAM) with multimodal deep learning models to improve transparency and decision support. A key limitation is that this framework remains conceptual and lacks extensive real world clinical validation across diverse healthcare settings.

Wang, L. et al.(2023) examines human-centered design and evaluation of AI-empowered clinical decision support systems (AI-CDSS) in healthcare settings. The authors analyze how AI-CDSS tools are developed, implemented, and assessed with a focus on clinician needs, usability, trust, and workflow integration. Findings show that while AI-CDSS can improve clinical efficiency and decision-making, successful adoption depends heavily on user experience factors such as transparency, usability, and alignment with clinical workflows. However, the review highlights persistent sociotechnical challenges, including workflow misalignment, usability issues, and lack of user trust. The study emphasizes that human-centered design is essential for effective real-world deployment of AI-CDSS.

Almulihi et al. (2024) conducted a systematic review to evaluate the application of artificial intelligence and machine learning in emergency department triage. The review analyzed 17 studies investigating AI and machine learning models for predicting patient outcomes and supporting triage decisions. Results showed that machine learning approaches generally outperformed conventional emergency severity scoring systems, with boosting algorithms demonstrating particularly strong performance. The study concluded that AI has significant potential to improve decision-making and patient prioritization in emergency departments. However, the authors emphasized the need for further validation and practical implementation studies before widespread clinical adoption.

Yi et al. (2025) conducted a systematic review of prospective studies to evaluate the effects of artificial intelligence-based triage in emergency departments. The review analyzed seven studies that examined outcomes related to AI-assisted triage systems and found that most approaches used machine learning models within five-level triage frameworks. Reported prediction accuracies ranged from approximately 80.5% to 99.1%, and several studies demonstrated improvements in triage efficiency, reductions in undertriage and overtriage, and better patient outcomes. The authors concluded that AI can serve as an effective support tool for triage nurses and emergency care providers. However, they emphasized that further prospective validation and broader clinical implementation studies are necessary before widespread adoption.

Proposed Solution:

This project proposes the development of a machine learning-based decision support system for emergency department triage. The system will be developed using a publicly available clinical dataset and will implement baseline classification models alongside an ensemble learning approach to predict patient triage levels. Model performance will be evaluated using standard classification metrics including precision, recall, and F1-score. Unlike existing studies that focus either on predictive performance or conceptual frameworks, this project focuses on the comparative model evaluation under controlled conditions to assess robustness and reliability in triage prediction. The scope of this work is limited to prototype development and offline evaluation and does not involve clinical deployment.

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